

Mochamad “Luthfan” F Akbar

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EDUCATION

CARNEGIE MELLON UNIVERSITY TEPPER SCHOOL OF BUSINESS

Pittsburgh, PA

Master of Business Administration – MBA Energy & Sustainability Track | Finance, Operations

5/26

- Relevant Coursework: Managerial & Global Economics, Optimization, Statistical Decision Making, Corporate Finance, Energy Finance, AI Methods for Data, Sustainable Finance, Probability & Statistics, Modern Data Management.
- Memberships: Graduate Finance Association, Adam Smith Society, Energy and Clean Tech.
- Teaching Experience:
 - Teaching Assistant – Finance I (2025) | Tutor – Probability & Statistics, Accounting, Operations (2025)
 - Teaching Assistant – Managerial Economics (2026)
- Awards & Honors: LPDP (Indonesia Endowment Fund for Education) Scholar.

INSTITUT TEKNOLOGI BANDUNG (ITB)

Bandung, Indonesia

Bachelor of Science in Petroleum Engineering

6/12

- Honors: SPE Student Paper Contest Finalist (“Optimization of Net Present Value Using Managed Water Loop Algorithm”) | National Doctoral Forum Presenter.
- Thesis: Published in ITB Digital Library and presented at SPE Forum (Beijing 2012).
- Memberships: Petroleum Engineers Student Association | Society of Petroleum Engineers.

RESEARCH EXPERIENCE

What Does a Green Hydrogen Mandate Cost? Evidence from the Intermountain Power Project

As Independent Researcher, 2026

- Built a techno-economic optimization model comparing grid-sourced, dedicated-renewable, and partial-mandate hydrogen production at the Intermountain Power Project, using two years of hourly CAISO SP-15 price data, site-calibrated solar/wind generation, and hourly grid carbon intensity, validated against the project’s own disclosed financials.
- Quantified the cost premium of a renewable-only supply mandate relative to grid-sourced supply, decomposed that premium into its underlying drivers (energy cost, asset utilization, and storage) to identify which lever matters most for cost reduction, and converted the finding into an implied carbon-abatement cost to benchmark the mandate against conventional carbon-pricing policy.
- Working paper posted on SSRN (DOI: 10.2139/ssrn.7428658); submitted to the International Journal of Hydrogen Energy, currently under peer review.

Levelized Cost of Storage (LCOS) Model for Zinc-Hybrid vs LFP Batteries (Capstone Project)

Carnegie Mellon University – Tepper School of Business, 2025

- Validated technical and economic parameters (RTE, DoD, degradation, augmentation, OPEX) using industry and second-life data.
- Developed a dynamic LCOS valuation model to compare zinc-bromide and lithium-ion (LFP) storage technologies.
- Linked technical performance to financial outcomes, supporting commercialization and investor-readiness analysis.

Gas Lift Conversion and Gas Flare Utilization Study

Pertamina Hulu Energi OSES, 2021 (Internal Study – Confidential)

- Conducted technical-economic assessment for converting electrical submersible pumps to gas lift and for reusing flare gas.
- The NPV-based analysis found the proposed pilot would be value-destructive under then-current costs, avoiding a negative-NPV capital commitment; facility and compressor designs were separately applied to the company’s flare-gas reduction program.

Iron Removal Agent for Polymer Flooding (Widuri Field)

CNOOC SES Ltd. & Starborn Chemical Company, 2015

- Evaluated chemical treatments to maintain polymer viscosity during enhanced oil recovery.
- Identified an effective additive combination (150 ppm S-9119, 100 ppm S-9363) to stabilize polymer injection.
- <https://www.ipa.or.id/en/publications/iron-removal-agent-for-polymer-flooding-project-of-widuri-c>

Optimization of Net Present Value Using Managed Water Loop Algorithm

Institut Teknologi Bandung, 2012

- Created simulation-based algorithms to manage downhole water flow and improve oil recovery efficiency.
- Evaluated economic impact using NPV comparison between conventional and downhole water loop wells, showing higher profitability under reservoir-pressure conservation conditions.
- Finalist, SPE Student Paper Contest; presented at National Doctoral Forum, Beijing.

WORK EXPERIENCE

PERTAMINA HULU ENERGI OSES – Indonesia National Oil and Gas Company

Jakarta, Indonesia

Artificial Lift Senior Supervisor – Production Optimization Engineer

9/18 – 7/24

- **Economic Evaluation & Project Valuation:** Performed investment analyses using DCF, NPV, and IRR models for multi-million-dollar well programs, including a \$30 million development-drilling campaign.
- **Feasibility Analysis:** Led the Artificial-Lift Conversion Study (ESP to Gas Lift with Flare Reuse), directing a cross-disciplinary team spanning reservoir, production, mechanical, and electrical engineering through an NPV-based investment case; the analysis found a proposed ~\$15 million pilot would be value-destructive, providing the basis to avoid a negative-NPV capital commitment.
- **Field Performance Optimization:** Applied data analytics and statistical monitoring (SCADA, Oracle, PowerBI) to evaluate 130 wells and reduce downtime across a portfolio of 165 active wells (\$85 million annual budget).
- **Strategic Project Management:** Coordinated multi-disciplinary teams (3 operator groups, 12 staff) in planning, logistics, and budget control for field-wide operations and maintenance programs.
- **Contract & Procurement Management:** Served on the Exploration Well Artificial-Lift Contract Team (\$1.5 million value), reviewing supplier proposals and ensuring technical and economic alignment with field plans.
- **Mentorship & Training:** Mentored junior engineers and student trainees, providing guidance on production analysis and cost modeling approaches.

CNOOC SES LTD. – China National Oil and Gas Company

Jakarta, Indonesia

Production/Petroleum Engineer

10/12 – 9/18

- **Strategic Project Management:** Served as acting Petroleum Engineer Specialist within the Management Development Program, contributing to long-term planning, project valuation, and economic performance reviews.
- **Performance Monitoring & Data Analytics:** Monitored 300 wells across 19 offshore platforms (\$115 million operations) using SCADA and PowerBI dashboards to inform budget allocation and production targets.
- **Project Management:** Managed well-work programs valued at \$0.5-2 million, overseeing budget forecasting, vendor coordination, and schedule compliance.
- **Research & Technology Implementation:** Led a \$9 million pilot Enhanced Oil Recovery (EOR) project, combining simulation and field testing to assess recovery improvement and economic feasibility.
- **Energy Systems Optimization:** Analyzed operations of a 90 MW field power system, identifying maintenance and fuel cost reduction options through process efficiency modelling.

SKILLS & CERTIFICATIONS

- **Quantitative & Economic Analysis:** Cost-benefit analysis | NPV/DCF/IRR modeling | Optimization Regression & sensitivity analysis | Policy evaluation methods
- **Data & Computing:** Python (NumPy, pandas, matplotlib) | PowerBI Microsoft Excel (advanced financial modeling) | SQL (data queries)
- **Energy & Sustainability Systems:** Energy transition economics | Hydrogen and storage modeling LCOS and LCOH evaluation | Electricity-market modeling | Project finance
- **Research & Communication:** Technical writing | Data visualization Teaching assistance (Finance I, Managerial Economics, Statistics) | Public presentation
- **Certifications:** (2023-2025)
Financial Statement, DCF & LBO Modeling – Wall Street Prep
Microeconomics: The Power of Markets & When Markets Fail – University of Pennsylvania
Mathematics for Machine Learning (Linear Algebra, PCA, Multivariate Calculus) – Imperial College London
Data Analytics & Visualization (Foundations, Data Cleaning, Statistical Methods) – Google
Human Resource Management and Leadership Specialization – Macquarie University

INTERESTS & AFFILIATIONS

- **Academic & Professional:** Society of Petroleum Engineers (SPE) | Graduate Finance Association (CMU Tepper) Adam Smith Society (CMU) | Energy and Clean Tech (CMU)
- **Research Interest:** Energy transition economics | Grid resilience | Sustainable infrastructure finance AI in energy systems | Industrial decarbonization policy
- **Personal Interests:** Hiking | Renewable project site visits | Strategic board games Mentorship and education outreach | Pro Bono Consulting